

The Probe and the Kernel

Resolving Power of a Fixed Reading Interface

Abstract

A projection that discards structure has not destroyed it, but neither has it automatically preserved it. This paper measures the difference. We take a fixed hexadecimal object, subject it to families of lawful single-symbol removals, and read the result through a non-aligned base transformation. The forward pass characterises what the transformation produces: a quadratic unfold law, a Benford leading-digit distribution, near-total decimal avalanche under single-symbol deletion, and a collision structure that occurs at exactly one class of event. The inverse pass asks what the response says about the source, and yields the paper's central result: **a probe resolves a source feature if and only if the walk sweeps that feature through the projection's kernel**. Measured across five source structures, two of three natural walks resolve 2 of 5; the third resolves 5 of 5; and the reason is derivable rather than empirical. We then apply the same observability condition to a second system — keyed superposition in a fixed-dimensional field — where the fiber is non-empty, and derive a closed-form capacity constant which is stated before measurement and confirmed at an untested dimension. The two systems differ in exactly one respect, and that respect determines which questions are askable of each.

1. Two Readings of One Object

The forward question and the inverse question are not competitors. They are the two halves of one measurement, and collapsing them is the error this paper spent the most effort correcting.

Forward. Given a fixed object and a specified family of lawful removals, what structure necessarily appears when the result is read across a base boundary? This asks what the transformation *produces*. It does not ask whether the production is the cheapest route to any particular fact.

Inverse. Given the produced response, what must have been true of the source to force it? This asks what the transformation *resolves*.

A feature being independently computable does not invalidate the forward pass. It bears only on the inverse pass, and there it is decisive. Both statements are used below and they are kept apart.

2. The Instrument

Object. A 64-symbol hexadecimal string \$H\$. The empty-string SHA-256 digest is used as the calibration sample throughout:

```
H = e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
N = int(H, 16) = 102987336249554097029535212322581322789799900648198034993379397001115665086549
```

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Operation. Lawful single-symbol removal, iterated. Three natural directions:

Walk	Removal index at length n
end	$n-1$
start	0
center	$\lfloor (n-1)/2 \rfloor$

Readout. The base transformation $\pi(H) = \text{int}(H, 16)$, applied at every step. The observable is the sequence, not any single value.

Derived observables. The change bit $\Delta_k \in \{0,1\}$ recording whether step k moved the integer; the decimal length-drop word; the leading-digit distribution; the shared-prefix fraction between consecutive rows.

3. Forward: What the Transformation Produces

3.1 The end and start walks are one identity

Measured across all 64 steps:

$$\text{end walk value at } k = N \div 16^k \quad \text{start walk value at } k = N \bmod 16^{64-k}$$

Both hold exactly. The two tables are the quotient and the remainder of one division, printed from opposite ends. Reconstruction of N from the pair at every k verifies.

Caution, recorded because it was violated in session: this identity appearing repeatedly across the work is not evidence of convergence between independent structures. Division appearing three times produces the split three times. One fact counted thrice is not a pattern. [Δ traversal artifact]

3.2 The unfold law is quadratic

A full one-at-a-time walk from n symbols emits rows of length $n-1, n-2, \dots, 1$. Total decimal symbols:

$$D(n) = \log_{10}(16) \cdot \frac{n^2}{2}$$

Input	Hex symbols	Decimal symbols out	Ratio
SHA-256 digest	64	2,460	31.9×
SHA-512 digest	128	9,852	63.9×
1 KB	2,048	2,525,015	1,023.9×
4 MB	8,388,608	42,366,205,509,364	4,194,269×

Quadratic in n , so the expansion **ratio** grows linearly with input size. A 4 MB object unfolds to 4.24 trillion decimal symbols under a single walk. [Ψ verified]

Arbitrary removal *order* is the larger space: $n!$ for a 4 MB object is $2^{180,835,781}$, exceeding the

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object's own $2^{\{33\}_{554}\{432\}}$ by $5.4\times$ in the exponent. Beyond $n = 44$ hex symbols there are more ordered readings of an object than there are possible objects of that size. [**Ψ verified**]

3.3 Single-symbol deletion is globally visible across the base boundary

Deleting one hex symbol and measuring displacement in the decimal representation, right-aligned, all 64 positions:

```
mean 0.8984   min 0.8289   max 0.9605
```

Reverse direction — delete one decimal symbol, measure hex displacement — gives mean 0.8106. The asymmetry is retained rather than averaged: the transformation is bijective at the integer level while local perturbations are **not** symmetric between coordinate systems.

Mechanism: $16 = 2^4$ is nibble-aligned to the binary lattice, so hex is location-locked. $10 = 2 \cdot 5$ has no clean place, so decimal is not. This is the same structural fact that forbids a base-10 digit-extraction spigot. **[Ψ verified]**

3.4 The length-drop word is Sturmian

Decimal length loses 1 or 2 symbols per hex symbol removed, never the average 1.204:

21112111121111211112111121111211112111121111211112111121111211112
13 twos of 64 = 0.203125 slope $\log_{10}(16) - 1 = 0.204120$

The word matches $\lfloor \log_{10} N - k \rfloor \log_{10} 16$ differences at $64/64$. The slope is universal across every 256-bit value; the object contributes only the phase $\frac{\log_{10} N}{\log_{10} 16}$.

Channel width, measured rather than assumed: sweeping all phases produces exactly **65 distinct length-64 drop words**, so the drop word distinguishes $\log_2 65 = 6.02$ bits of a 256-bit object. The pattern is real without being a high-capacity channel. [**Ψ verified, scoped**]

3.5 Collisions occur at exactly one class of event

Object	end	start	center	pooled
plain digest	64/64 distinct	63/64	64/64	188/192
20-zero-prefixed	51/70	50/70	50/70	128/210

On an object with no zero field, the value sequence is essentially collision-free — 188 distinct values from 192 rows across three walks. Every collision that does occur traces to a leading-zero event, since `$\operatorname{int}$` is blind to leading zeros by construction. The single collision in the plain digest's start walk is the 0 in `e3bo`, caught the moment it reaches the front.

Collisions in the value sequence occur at zero-plateau events and nowhere else. This is derivable from the kernel of $\$ \backslash \pi \$$, not an empirical regularity. [Ψ verified]

3.6 Regularity is chosen by direction

Walk	Successive ratio range
end	16.000000 ... 16.214286

Walk	Successive ratio range
center	15.861815 ... 45.800000

All three walks are monotone decreasing with zero increases. But the end walk is a near-clean geometric ladder while the center walk's ratio varies by a factor of three. Same object, same readout — the progression is regular or irregular according to which direction the reader takes. [**Ψ verified**]

3.7 Benford structure inside the scatter

Leading decimal digits across center-walk rows:

1:18 2:11 3:8 4:6 5:6 6:5 7:3 8:3 9:4
 $18/64 = 0.281$ Benford $P(1) = 0.301$

And the digits are genuinely scattered: mean shared decimal prefix between consecutive rows is **0.0027** — for the end walk as well as the center walk. Even the regular $\div 16$ ladder rewrites nearly every decimal symbol at each step.

So the sequence is simultaneously scattered at the symbol level and law-abiding at the leading-digit level. Both are consequences of a geometric-ish sequence being equidistributed in $\log \bmod 1$. [**Ψ verified**]

Correction recorded: a chi-square test on the pooled digit stream returned $\chi^2 = 38.09$ and was reported as non-uniformity. The test is invalid — the rows are nested truncations differing by a factor of 16 and are nowhere near independent. Withdrawn. [**⊥ design failure**]

4. Inverse: What the Response Resolves

4.1 The resolving test

Five source structures, all length 70, same fixed 50-symbol data payload. Observable: the change-bit signature per walk.

Source structure	end	center	start
leading zeros (20)	visible	visible	visible
middle zeros (20)	blind	blind	visible
trailing zeros (20)	blind	blind	visible
leading f-run (20)	blind	blind	blind
no special structure	blind	blind	visible
distinct signatures	2/5	2/5	5/5

Combining all three change-bit signatures with the length-drop word: **5/5 distinct, zero collisions.**

4.2 The kernel-sweep rule

The kernel of $\pi = \int (\cdot, 16)$ is exactly *leading zeros*. The start walk is the only

direction that marches every position through the leading slot. Every zero anywhere in the object therefore eventually becomes a leading zero and registers as a plateau. The end and center walks never sweep anything through the kernel, so they detect only what already sat in it.

A probe resolves a source feature if and only if the walk moves that feature through the projection's kernel.

This is a design rule, not a metaphor. It determines which walk to construct for a given readout, rather than leaving the choice to preference. It also explains why the leading-zero case appeared so vivid: it is the one feature that begins inside the kernel, so *every* walk catches it. Generalising from that sample is generalising from the easiest possible case. **[Ψ theorem, within this projection class]**

Predictions logged and failed: trailing zeros were predicted invisible to all three walks — the start walk resolves them cleanly. Middle zeros were predicted visible to the center walk — it is blind to them. Both wrong, and the kernel-sweep rule was extracted from the failures. **[$\perp$ two predictions falsified]**

4.3 The walks are unequal instruments

End and center are 2/5 instruments on this readout. They are not equivalent alternative directions; they are substantially weaker probes, and the weakness is derivable in advance from the kernel. Route independence works here as case-set spanning — the three walks together resolve what none resolves alone — but not as method diversity for its own sake. **[Ψ verified]**

5. The Observability Condition

The forward and inverse passes require different preconditions, and this section states the inverse one exactly.

Let x be the hidden state, π the available projection, m all retained metadata, and T a lawful probe. The static observer holds $O_o(x) = (\pi(x), m(x))$. The dynamic observer holds the orbit $O_T(x) = (O_o(x), O_o(Tx), O_o(T^2x), \dots)$.

The probe has genuine resolving power only if there exist $x \neq x'$ with $O_o(x) = O_o(x')$ but $O_T(x) \neq O_T(x')$.

This is the observability condition. It is not new — two internal states in one observation class, separated by their forward orbits, is Kalman's 1960 criterion and the founding condition of control theory. It is stated here because applying it retroactively splits the work cleanly.

5.1 The hex object has an empty fiber

For a string of the form $o^z X$ with X canonical:

$$z = L - \ell_{16}(\pi(H))$$

so $(\pi(H), L)$ determines H outright. Verified: reconstruction of the full string from (value, length) succeeds exactly.

The fiber is empty. No x' shares $O_o(x)$. No probe can separate anything, because nothing is unseparated. Every walk, every unfold, every re-cast is a function of what the observer already held.

This has a consequence sharper than any counting argument: the reason arbitrary data cannot be stored in and retrieved from a fixed digest is not primarily that the address costs more than the content. It is that **there is no hidden state to address**. The counting result is the symptom; this is the cause.

Scope note. This kills a specific epistemic claim — that the walk recovers information unavailable to the static observer — and kills nothing about §3. The forward pass characterises what the transformation produces and is untouched by the fiber being empty. A calibration run deliberately uses samples whose properties are known; redundancy there is the requirement, not the defect. The two were conflated in session and the conflation is corrected here. [**⊥ epistemic claim falsified; Ψ forward pass intact**]

5.2 A keyed superposition field has a large fiber

Second system. C codewords in $\{-1, +1\}^D$; m of them bound to independent keys by elementwise product and summed into one field S .

Observation	Result
static read of S against the codebook, top- m	0/4, 0/4 (chance 0.25)
probe read, unbind by key	4/4, 4/4

S does not determine the item set. The key is a probe, not metadata — it is not derivable from S or the codebook, and nothing sitting alongside gives it away. **Condition satisfied.**

Extended across load:

m	head-on hit rate	probe hit rate	chance
1	0.0175	1.0000	0.0156
2	0.0138	1.0000	0.0156
4	0.0156	1.0000	0.0156
8	0.0159	0.9988	0.0156
16	0.0144	0.9470	0.0156

Head-on sits at chance from $m=1$. Not slower — **blind**. The payload is orthogonal to the field containing it, so direct inspection returns nothing at any budget. One elementwise product and it is fully present.

Unreadable on-axis at infinite budget; readable immediately off-axis. [**Ψ verified**]

5.3 Identification is joint

Holding $m = 6$ and varying field dimension against admissible-set size:

$ C $	$D=64$	$D=128$	$D=256$	$D=512$	$D=1024$
16	0.940	0.997	1.000	1.000	1.000
64	0.829	0.993	1.000	1.000	1.000
256	0.713	0.977	0.999	1.000	1.000
1024	0.536	0.944	0.999	1.000	1.000

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Both axes move recovery. The payload is located in neither the field nor the admissible set. Cue plus constraint equals identification, and the relation is the location.

Correction: the earlier statement "the payload is not in the field, it is in the admissible set" is too strong and is withdrawn. [1 falsified]

6. Capacity of a Fixed Interface

6.1 The derivation

After unbinding with K_1 : $R = P_1 + \sum_{i \geq 2} K_1 K_i P_i$. Correlating against the codebook gives correct score $\sim \mathcal{N}(D, (m-1)D)$ and $C-1$ competitor scores $\sim \mathcal{N}(0, mD)$. Identification requires the correct score to exceed the maximum competitor. Dividing by $\sqrt{D}$ in the large- m limit:

$$m_{\max} = \alpha D, \quad \alpha = \frac{1}{\left(\mathbb{E}[\max_{C-1}] + z_{\epsilon}\right)^2}$$

$\ C\ $	$\mathbb{E}[\max_{C-1}]$	α
16	1.5694	0.06589
64	2.1921	0.04898
256	2.6917	0.03971
1024	3.1231	0.03367

Against eighteen measured cells spanning $16 \times$ in dimension and $64 \times$ in codebook size: **mean measured/predicted 0.891, sd 0.056**, nothing fitted. The residual is the two-term extreme-value expansion converging slowly.

6.2 Prediction and test

Stated before running, at an untested dimension:

	predicted (raw)	predicted ($\times 0.891$)	measured	counting bound
$D=2048, \ C\ =256$	81.3	72	70	256
$D=2048, \ C\ =1024$	69.0	61	62	205

Within 3% and 2%. The counting bound is wrong by $3.7 \times$ and $3.3 \times$; the derivation is right to within its known bias.

The gap between abstract state-space capacity and capacity accessible through one fixed reading interface is not a resource limitation that scale closes. It is a fixed multiplicative price of the algebra, and it has a closed form. [Ψ theorem]

6.3 Component decomposition

At $D = 512$, $|C| = 256$:

Variant	$m_{\max}$
baseline (dense random code, raw sum, NN cleanup)	18
bundle by sign (1-bit field)	11
max-distance code (Hadamard rows)	18

Codebook geometry contributes nothing — Hadamard rows are exactly orthogonal, random codes only near-orthogonal, identical capacity. $d_{\min}$ is removed from the candidate terms. Quantising the field to one bit costs 39%: the binary interface over an analog interior is priced, and the price is about two fifths of the load. [**Ψ verified; one null**]

Withdrawn: a sparse-payload variant returned $m_{\max} = 2$. The implementation masked the bound vector, discarding 90% of the payload. No result either way. [**⊥ design failure**]

6.4 Substrate independence and its limit

Algebra	$D=512, C =256$	$D=1024, C =64$
elementwise multiply	0.93	0.84
cyclic shift, distinct keys	0.89	0.84
full random permutation	0.89	0.80
circular convolution (HRR)	0.84	0.80
Gaussian payload alphabet	0.93	0.82

One law, five substrates. A sixth case — cyclic shift with keys drawn *with* replacement — returned 0.49 and 0.40, and the failure was a birthday collision in the key space at $m \sim \sqrt{D}$, not a failure of the law. Restoring key distinctness restores the law exactly.

The binding requirement is key distinctness, not key-space cardinality. D distinct keys perform identically to $D!$ keys. This corrects a stated prediction that a size- D key space would give only partial recovery. [**Ψ verified; one prediction falsified productively**]

Scope caveat, recorded rather than omitted: all five substrates are group actions. Transfer across them is weaker evidence than it appears, in the same way that Z80 and ARM are both von Neumann machines. A genuine test requires a substrate with no invertible key operation — Hopfield storage at capacity $0.138N$, derived by an unrelated route, is the obvious candidate and is not attempted here. [**Ω open**]

7. What Cannot Be Stored

The forward pass produces a $31.9\times$ symbol expansion from a 64-symbol digest under one walk. The question of whether chosen data can be placed there is settled by two independent measurements.

Reachability. Every possible digest crossed with every ordered walk over it addresses 2^{552} outputs. Distinct 5 MB files number $2^{41,943,040}$. Fraction reachable: $2^{-41,942,488}$. Not expensive to find — not in the image of the map.

Search cost. Choosing a target first and hunting for a digest that produces it, measured over 2×10^6 trials:

10 hashes -> 2 digits matched	100,000 -> 5
1,000 -> 3	1,000,000 -> 6
10,000 -> 4	

Clean factor of ten per matched digit. Row 1 alone is 76 digits, requiring 10^{76} trials; the full 63-row table is 2,454 digits.

The same curve, priced by a market. Bitcoin mining is this search. At ~1 ZH/s the network performs 6.0×10^{23} hashes per block and 3.2×10^{28} per year — **23.8 digits per block, 28.5 digits per year**, against 2,454 needed. Seventeen years of purpose-built silicon and gigawatt-scale power confirm the slope measured on one laptop.

Where the bits actually go. The architecture the forward pass gestures at exists and ships. A diffusion latent of 64 KB unfolds into a full image through a 1.72 GB decoder — 26,245x. A Bitcoin block header is 80 bytes and commits to a ~700 GB chain. In both cases the small object addresses and the large object holds. Data has entropy H ; the split between seed and decoder is a free design axis and the sum is not. **[Ψ verified]**

7.1 Proof-of-work is invisible to the value and visible in the sequence

Genesis block, 10 leading zeros: the start walk produces **ten consecutive identical values**, then an avalanche at the eleventh. The decimal representation loses 13 symbols. At modern difficulty, 76 bits of proof-of-work — the only part of a mined digest that costs anything — reduce the decimal output from 78 to 53 symbols.

Every zero bought shortens the unfold. A maximally-mined digest is a maximally-poor seed for this walk. **[Ψ verified]**

Recorded: the zero count is also $2^{56} - \log_2(H)$, one operation. The walk is a slow route to a fast fact, which bears on §5 and not on §3.

8. Status Ledger

ID	Statement	Status
F1	End/start walks are the quotient and remainder of one division	Ψ verified
F2	Unfold law $D(n) = \log_{10}(16)n^{2/2}$; ratio linear in n	Ψ verified
F3	Single-symbol deletion displaces 0.8289–0.9605 of decimal symbols	Ψ verified
F4	Length-drop word is Sturmian; 65	Ψ verified, scoped

ID	Statement	Status
	distinct words; 6.02 bits	
F5	Value collisions occur at zero-plateau events and nowhere else	Ψ verified
F6	Regularity of the progression is selected by walk direction	Ψ verified
F7	Leading digits Benford; shared prefix 0.0027 in both walks	Ψ verified
I1	Kernel-sweep rule: a probe resolves iff the walk sweeps the kernel	Ψ theorem (this projection class)
I2	End and center walks resolve 2/5; start resolves 5/5; combined 5/5	Ψ verified
O1	Observability condition as the precondition for inverse recovery	Ψ (Kalman 1960; restated)
O2	The hex object's fiber under (value, length) is empty	Ψ verified
O3	Keyed superposition has a large fiber; head-on blind at all \$m\$	Ψ verified
O4	Identification is joint in field and admissible set	Ψ verified
C1	$\alpha = 1/(\mathbb{E}[\max_{\{C-1\}} + z_{\epsilon}]^2)$; 18 cells at 0.891 ± 0.056	Ψ theorem
C2	Forward prediction at $D=2048$: $81.3/69.0$ stated, $70/62$ measured	Ψ verified
C3	Codebook geometry contributes nothing; Hadamard $\equiv$ random	Ψ null
C4	1-bit field quantisation costs 39% of load	Ψ verified
C5	Binding requires key distinctness, not key-space cardinality	Ψ verified
S1	Arbitrary target unreachable: $2^{\{-41,942,488\}}$ of 5 MB files	Ψ verified
S2	Reverse search costs 10x per matched digit; Bitcoin confirms at scale	Ψ verified
X1	Walk recovers zero count unavailable to the static observer	⊥ falsified (metadata-cheap)
X2	Payload is located in the	⊥ falsified (joint)

ID	Statement	Status
	admissible set	
X ₃	Key-key correlation governs retrieval	⊥ design failure (two mechanisms, one value)
X ₄	Sparse payloads collapse capacity to $m=2$	⊥ design failure (masked the bound vector)
X ₅	Drop word carries ~12.7 bytes of constraint	⊥ falsified (6.02 bits)
X ₆	Digit stream is non-uniform, $\chi^2 = 38.09$	⊥ design failure (nested rows)
X ₇	Trailing zeros invisible to all walks	⊥ falsified (start resolves)
X ₈	Center walk resolves middle zeros	⊥ falsified (blind)
A ₁	Winding/phase split recurring as evidence of convergence	Δ traversal artifact
A ₂	Five-substrate transfer as strong independence evidence	Δ all five are group actions
Ω ₁	A projection whose kernel is not metadata-recoverable	Ω open — BBP annihilator is the candidate
Ω ₂	Capacity law across a non-group substrate (Hopfield, 0.138N)	Ω open
Ω ₃	Kernel-sweep rule beyond single-symbol removal families	Ω open

Ψ = verified · ⊥ = falsified or design failure · Δ = artifact of the reader's traversal · Ω = open

9. The Sharp Open Edge

Ω₁ is the one that matters, and §5 says why. The hex object cannot host an inverse result because its fiber under retained observation is empty — the instrument was calibrated on a sample with nothing hidden in it, which is correct practice for calibration and useless as a destination.

The BBP annihilator lattice has the required shape. Its kernel dimension is not advertised by the coefficient vector, there is no length coordinate that hands it over, and the existing frame table contains cells where the reader gap $\dim R - \gcd(C, q)$ is nonzero without an accounted mechanism. That is a fiber with something in it.

The instrument built here — lawful removal, non-aligned readout, response-surface comparison, and the kernel-sweep rule that says which walk to construct — is what should be pointed at it.

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measurement output. Nothing is estimated or recalled.